

Youbin He

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Education

 **The Hong Kong Polytechnic University**

Hong Kong | Sep. 2025 – Jun. 2029 (Expected)

 **BEng (Hons) in Aviation Engineering, minor in Artificial Intelligence & Data Analytics (AIDA)**

Publications

Youbin He, Siwei Wang. P-SRM: Selective Recovery of Rejected Predictions in Visual Tracking.

Manuscript submitted to ICASSP 2027; under review.

[arXiv](#) | [GitHub](#)

Projects

P-SRM: Lightweight Post-Rejection Selective Recovery for Visual Tracking

Research project

- **Problem formulation:** While reproducing TrackNetV3, TAPIR, KCF, and other trackers, identified correctly localized candidates discarded by native rejection. Proposed P-SRM through candidate-error categorization and analysis of high-scoring false acceptances.
- **Method design:** Built a unified evidence adapter encoding response maps, candidate support, and valid regions as $3 \times 32 \times 32$ tensors. Combined these with candidate geometry in a CNN with approximately 1.23M parameters, using auxiliary target-presence supervision, pAUC optimization, native rejection margins, and motion history for selective recovery while preserving native accepted outputs and candidate coordinates.
- **Evaluation:** Ran three-seed experiments across six methods, four datasets, and nine configurations spanning three tracking tasks. Achieved maximum gains of 1.27 percentage points in pooled F1 and 32.13 points in AP_r over native baselines; assessed component contributions through ablations and additional inference cost through timing measurements.

HomeSafe: Action Safety Analysis for Household Robots under Limited Perception

Personal project

- **Problem formulation:** Designed a three-way authorization framework—allow, deny, or insufficient evidence—for household robot actions under limited perception. Paired perception configurations with decision rules to evaluate dangerous authorizations, excessive stopping, and unresolved decisions.
- **System implementation:** Built an OmniGibson environment on NVIDIA Isaac Sim using BEHAVIOR-1K and a static robot model from the official Go2 URDF and digital-asset geometry. Established world–body–sensor transforms; configured RGB, linear depth, and ray-based simulated 2D ranging; integrated YOLO-World detection, action-region projection, and independent geometric collision queries for perception and reference judgments.
- **Development results:** Collected 80 development states, producing 21,501 detections and 400 state–action records. Used visualizations to identify near-field coverage and camera-extrinsic issues, revised the camera to a 120° field of view and 40° downward tilt, and completed local validation and multimodal report generation for the revised pipeline.

Experience

 **Lens Intelligent Robot (Changsha) Co., Ltd. — Intern**

Changsha | Jun. – Sep. 2026

- **Design & validation:** Contributed to 6-DOF hand development and design input for the ongoing 22-DOF hand. Used motion analysis, interference checks, and iterative 3D printing to refine structures, removing approximately 20.31 mm^3 of interference. Assembled prototypes, tested homing and binding, and supported motor and embedded control-software validation.
- **Manufacturing & debugging:** Supported BrainCo 13/16-DOF hand manufacturing; organized BOMs and SAP material records for eight configurations and redrew more than 40 engineering drawings. Delivered assembly training, supported validation of six prototype pairs, and assisted with RS-485 configuration, cable soldering, and test-software debugging.
- **Embodied AI deployment & evaluation:** Deployed Gemini 2 depth cameras on humanoid and quadruped robots and integrated OpenVLA. Completed end-to-end training and testing for tasks spanning locomotion and object grasping, and analyzed results across seven scene categories in the test environment.
- **Client demonstration:** Configured the SDK and PySide6 interface for a 21-DOF hand demonstration to a visiting NVIDIA team. Resolved serial-port and connection issues and explained DOF configuration, tactile sensing, and MIT impedance-control-based finger actuation in English.

Skills

Programming & Machine Learning: Python, TypeScript, PyTorch, NumPy, OpenCV, Git, Shell, Linux/WSL2, Conda environments.

Mechanical Design & Hardware: SolidWorks, Creo, part and board-outline drawings, 3D-printed prototyping, assembly testing.

Languages: Chinese (native); English (IELTS 6.5, 2024).